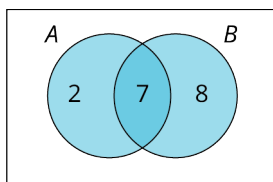
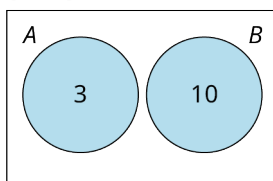


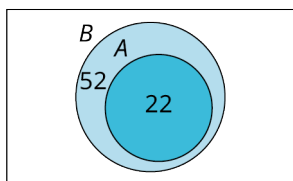
47. $U = 21$



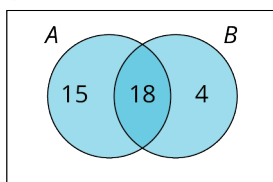
48. $U = 15$



49. $U = 88$



50. $U = 44$



1.5 Set Operations with Three Sets

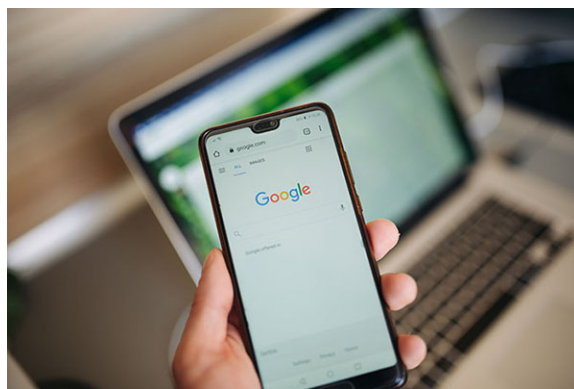


Figure 1.28 Companies like Google collect data on how you use their services, but the data requires analysis to really mean something. (credit: “Man holding smartphone and searches through google” by Nenad Stojkovic/Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Interpret Venn diagrams with three sets.
2. Create Venn diagrams with three sets.
3. Apply set operations to three sets.
4. Prove equality of sets using Venn diagrams.

Have you ever searched for something on the Internet and then soon after started seeing multiple advertisements for that item while browsing other web pages? Large corporations have built their business on data collection and analysis. As we start working with larger data sets, the analysis becomes more complex. In this section, we will extend our knowledge of set relationships by including a third set.

A Venn diagram with two intersecting sets breaks up the universal set into four regions; simply adding one additional set will increase the number of regions to eight, doubling the complexity of the problem.

Venn Diagrams with Three Sets

Below is a Venn diagram with two intersecting sets, which breaks the universal set up into four distinct regions.

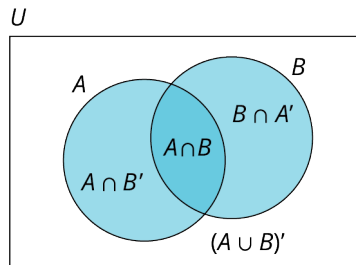


Figure 1.29

Next, we see a **Venn diagram with three intersecting sets**, which breaks up the universal set into eight distinct regions.

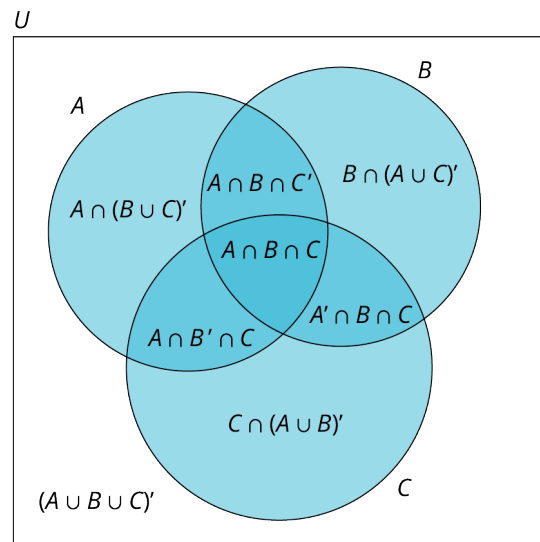


Figure 1.30

TECH CHECK

Shading Venn Diagrams

Venn Diagram is an Android application that allows you to visualize how the sets are related in a Venn diagram by entering expressions and displaying the resulting Venn diagram of the set shaded in gray.

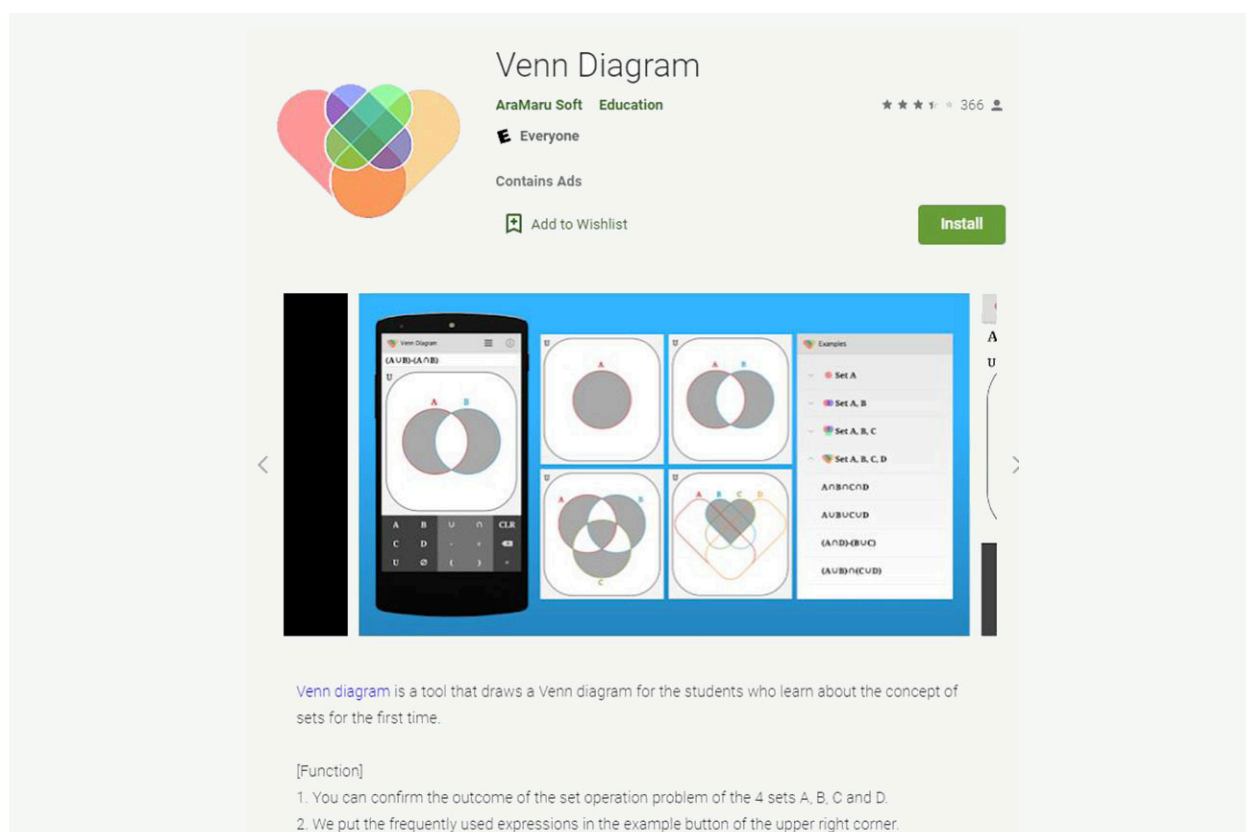


Figure 1.31 Google Play Store image of Venn Diagram app. (credit: screenshot from Google Play)

The Venn Diagram application uses some notation that differs from the notation covered in this text.

- The complement of set A in this text is written symbolically as A' , but the Venn Diagram app uses A^C to represent the complement operation.
- The set difference operation, $-$, is available in the Venn Diagram app, although this operation is not covered in the text.

It is recommended that you explore this application to expand your knowledge of Venn diagrams prior to continuing with the next example.

In the next example, we will explore the three main blood factors, A, B and Rh. The following background information about blood types will help explain the relationships between the sets of blood factors. If an individual has blood factor A or B, those will be included in their blood type. The Rh factor is indicated with a $+$ or a $-$. For example, if a person has all three blood factors, then their blood type would be AB^+ . In the Venn diagram, they would be in the intersection of all three sets, $A \cap B \cap Rh^+$. If a person did not have any of these three blood factors, then their blood type would be O^- , and they would be in the set $(A \cup B \cup Rh^+)^c$ which is the region outside all three circles.

EXAMPLE 1.35

Interpreting a Venn Diagram with Three Sets

Use the Venn diagram below, which shows the blood types of 100 people who donated blood at a local clinic, to answer the following questions.

U = Blood types of 100 People

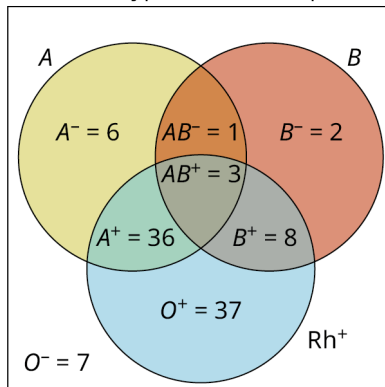


Figure 1.32

1. How many people with a type A blood factor donated blood?
2. Julio has blood type B^+ . If he needs to have surgery that requires a blood transfusion, he can accept blood from anyone who does not have a type A blood factor. How many people donated blood that Julio can accept?
3. How many people who donated blood do not have the Rh^+ blood factor?
4. How many people had type A and type B blood?

✓ Solution

1. The number of people who donated blood with a type A blood factor will include the sum of all the values included in the A circle. It will be the union of sets A^- , A^+ , AB^- and AB^+ .

$$n(A) = n(A^-) + n(A^+) + n(AB^-) + n(AB^+) = 6 + 36 + 1 + 3 = 46.$$
2. In part 1, it was determined that the number of donors with a type A blood factor is 46. To determine the number of people who did not have a type A blood factor, use the following property, A' union is equal to U , which means $n(A) + n(A') = n(U)$, and $n(A') = n(U) - n(A) = 100 - 46 = 54$. Thus, 54 people donated blood that Julio can accept.
3. This would be everyone outside the Rh^+ circle, or everyone with a negative Rh factor,

$$n(Rh^-) = n(O^-) + n(A^-) + n(AB^-) + n(B^-) = 7 + 6 + 1 + 2 = 16.$$
4. To have both blood type A and blood type B, a person would need to be in the intersection of sets A and B . The two circles overlap in the regions labeled AB^- and AB^+ . Add up the number of people in these two regions to get the total: $1 + 3 = 4$. This can be written symbolically as $n(A \text{ and } B) = n(A \cap B) = n(AB^-) + n(AB^+) = 1 + 3 = 4$.

> YOUR TURN 1.35

Use the same Venn diagram in the example above to answer the following questions.

1. How many people donated blood with a type B blood factor?
2. How many people who donated blood did not have a type B blood factor?
3. How many people who donated blood had a type B blood factor or were Rh^+ ?

?? WHO KNEW?

Blood Types

Most people know their main blood type of A, B, AB, or O and whether they are Rh^+ or Rh^- , but did you know that the International Society of Blood Transfusion recognizes twenty-eight additional blood types that have important implications for organ transplants and successful pregnancy? For more information, check out this article:

[Blood mystery solved: Two new blood types identified \(https://openstax.org/r/Two-new-blood-types-identified\)](https://openstax.org/r/Two-new-blood-types-identified)

Creating Venn Diagrams with Three Sets

In general, when creating Venn diagrams from data involving three subsets of a universal set, the strategy is to work

from the inside out. Start with the intersection of the three sets, then address the regions that involve the intersection of two sets. Next, complete the regions that involve a single set, and finally address the region in the universal set that does not intersect with any of the three sets. This method can be extended to any number of sets. The key is to start with the region involving the most overlap, working your way from the center out.

EXAMPLE 1.36

Creating a Venn Diagram with Three Sets

A teacher surveyed her class of 43 students to find out how they prepared for their last test. She found that 24 students made flash cards, 14 studied their notes, and 27 completed the review assignment. Of the entire class of 43 students, 12 completed the review and made flash cards, nine completed the review and studied their notes, and seven made flash cards and studied their notes, while only five students completed all three of these tasks. The remaining students did not do any of these tasks. Create a Venn diagram with subsets labeled: “Notes,” “Flash Cards,” and “Review” to represent how the students prepared for the test.

Solution

Step 1: First, draw a Venn diagram with three intersecting circles to represent the three intersecting sets: Notes, Flash Cards, and Review. Label the universal set with the cardinality of the class.

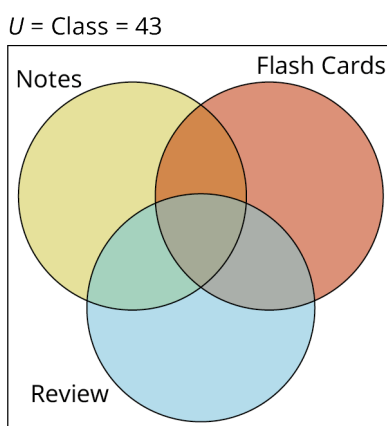


Figure 1.33

Step 2: Next, in the region where all three sets intersect, enter the number of students who completed all three tasks.

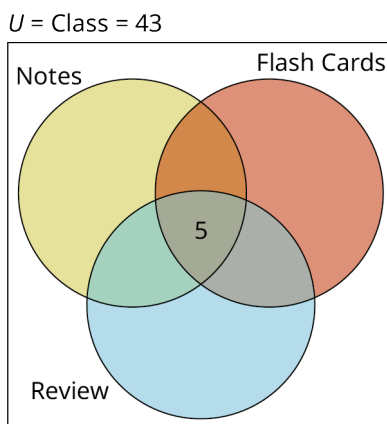


Figure 1.34

Step 3: Next, calculate the value and label the three sections where just two sets overlap.

- Review and flash card overlap.** A total of 12 students completed the review and made flash cards, but five of these twelve students did all three tasks, so we need to subtract: $12 - 5 = 7$. This is the value for the region where the flash card set intersects with the review set.
- Review and notes overlap.** A total of 9 students completed the review and studied their notes, but again, five of these nine students completed all three tasks. So, we subtract: $9 - 5 = 4$. This is the value for the region where the review set intersects with the notes set.

- c. **Flash card and notes overlap.** A total of 7 students made flash cards and studied their notes; subtracting the five students that did all three tasks from this number leaves 2 students who only studied their notes and made flash cards. Add these values to the Venn diagram.

$$U = \text{Class} = 43$$

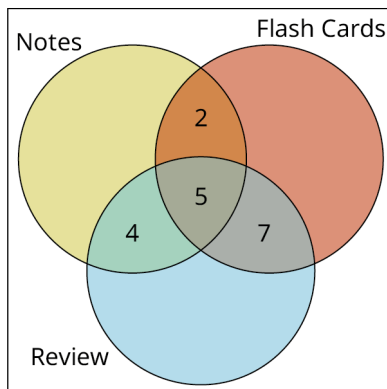


Figure 1.35

Step 4: Now, repeat this process to find the number of students who only completed one of these three tasks.

- A total of 24 students completed flash cards, but we have already accounted for $2 + 5 + 7 = 14$ of these. Thus, $24 - 14 = 10$ students who just made flash cards.
- A total of 14 students studied their notes, but we have already accounted for $4 + 5 + 2 = 11$ of these. Thus, $14 - 11 = 3$ students only studied their notes.
- A total of 27 students completed the review assignment, but we have already accounted for $4 + 5 + 7 = 16$ of these, which means $27 - 16 = 11$ students only completed the review assignment.
- Add these values to the Venn diagram.

$$U = \text{Class} = 43$$

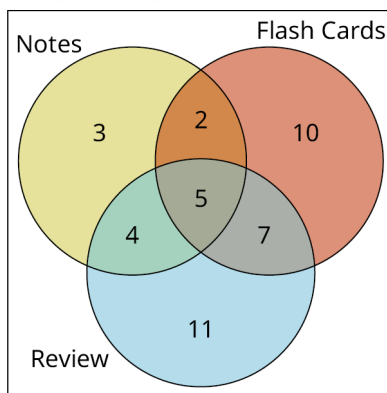


Figure 1.36

Step 5: Finally, compute how many students did not do any of these three tasks. To do this, we add together each value that we have already calculated for the separate and intersecting sections of our three sets:

$3 + 2 + 4 + 5 + 10 + 7 + 11 = 42$. Because there are 43 students in the class, and $43 - 42 = 1$, this means only one student did not complete any of these tasks to prepare for the test. Record this value somewhere in the rectangle, but outside of all the circles, to complete the Venn diagram.

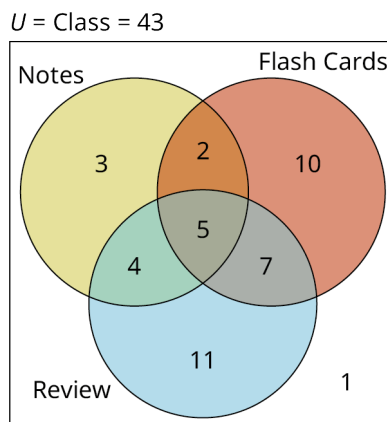


Figure 1.37

YOUR TURN 1.36

1. A group of 50 people attending a conference who preordered their lunch were able to select their choice of soup, salad, or sandwich. A total of 17 people selected soup, 29 people selected salad and 35 people selected a sandwich. Of these orders, 11 attendees selected soup and salad, 10 attendees selected soup and a sandwich, and 18 selected a salad and a sandwich, while eight people selected a soup, a salad, and a sandwich. Create a Venn diagram with subsets labeled "Soup," "Salad," and "Sandwich," and label the cardinality of each section of the Venn diagram as indicated by the data.

Applying Set Operations to Three Sets

Set operations are applied between two sets at a time. Parentheses indicate which operation should be performed first. As with numbers, the inner most parentheses are applied first. Next, find the complement of any sets, then perform any union or intersections that remain.

EXAMPLE 1.37

Applying Set Operations to Three Sets

Perform the set operations as indicated on the following sets: $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$, $A = \{0, 1, 2, 3, 4, 5, 6\}$, $B = \{0, 2, 4, 6, 8, 10, 12\}$, and $C = \{0, 3, 6, 9, 12\}$.

1. Find $(A \cap B) \cap C$.
2. Find $A \cap (B \cup C)$.
3. Find $(A \cap B) \cup C'$.

Solution

1. Parentheses first, A intersection B equals $A \cap B = \{0, 2, 4, 6\}$, the elements common to both A and B . $(A \cap B) \cap C = \{0, 2, 4, 6\} \cap \{0, 3, 6, 9, 12\} = \{0, 6\}$, because the only elements that are in both sets are 0 and 6.
2. Parentheses first, B union C equals $B \cup C = \{0, 2, 3, 4, 6, 8, 9, 10, 12\}$, the collection of all elements in set B or set C or both. $A \cap (B \cup C) = \{0, 1, 2, 3, 4, 5, 6\} \cap \{0, 2, 3, 4, 6, 8, 9, 10, 12\} = \{0, 2, 3, 4, 6\}$, because the intersection of these two sets is the set of elements that are common to both sets.
3. Parentheses first, A intersection B equals $A \cap B = \{0, 2, 4, 6\}$. Next, find C' . The complement of set C is the set of elements in the universal set U that are not in set C . $C' = \{1, 2, 4, 5, 7, 8, 10, 11\}$. Finally, find $(A \cap B) \cup C' = \{0, 2, 4, 6\} \cup \{1, 2, 4, 5, 7, 8, 10, 11\} = \{0, 1, 2, 4, 5, 6, 7, 8, 10, 11\}$.

YOUR TURN 1.37

Using the same sets from Example 1.37, perform the set operations indicated.

1. Find $A \cap (B \cap C)$.
2. Find $(A \cap B) \cup (A \cap C)$.
3. Find $(A \cup C') \cap (B \cup C')$.

Notice that the answers to the Your Turn are the same as those in the Example. This is not a coincidence. The following equivalences hold true for sets:

- $A \cap (B \cap C) = (A \cap B) \cap C$ and $A \cup (B \cup C) = (A \cup B) \cup C$. These are the associative property for set intersection and set union.
- $A \cap B = B \cap A$ and $A \cup B = B \cup A$. These are the commutative property for set intersection and set union.
- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$. These are the distributive property for sets over union and intersection, respectively.

Proving Equality of Sets Using Venn Diagrams

To prove set equality using Venn diagrams, the strategy is to draw a Venn diagram to represent each side of the equality, then look at the resulting diagrams to see if the regions under consideration are identical.

Augustus De Morgan was an English mathematician known for his contributions to set theory and logic. De Morgan's law for set complement over union states that $(A \cup B)' = A' \cap B'$. In the next example, we will use Venn diagrams to prove De Morgan's law for set complement over union is true. But before we begin, let us confirm De Morgan's law works for a specific example. While showing something is true for one specific example is not a proof, it will provide us with some reason to believe that it may be true for all cases.

Let $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 3, 4\}$, and $B = \{3, 4, 5, 6\}$. We will use these sets in the equation $(A \cup B)' = A' \cap B'$. To begin, find the value of the set defined by each side of the equation.

Step 1: $A \cup B$ is the collection of all unique elements in set A or set B or both. $A \cup B = \{2, 3, 4, 5, 6\}$. The complement of A union B , $(A \cup B)'$, is the set of all elements in the universal set that are not in $A \cup B$. So, the left side the equation $(A \cup B)'$ is equal to the set $\{1, 7\}$.

Step 2: The right side of the equation is $A' \cap B'$. A' is the set of all members of the universal set U that are not in set A . $A' = \{1, 5, 6, 7\}$. Similarly, $B' = \{1, 2, 7\}$.

Step 3: Finally, $A' \cap B'$ is the set of all elements that are in both A' and B' . The numbers 1 and 7 are common to both sets, therefore, $A' \cap B' = \{1, 7\}$. Because, $\{1, 7\} = \{1, 7\}$ we have demonstrated that De Morgan's law for set complement over union works for this particular example. The Venn diagram below depicts this relationship.

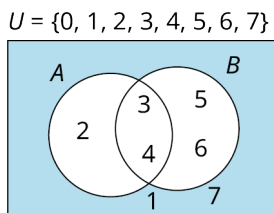


Figure 1.38

EXAMPLE 1.38

Proving De Morgan's Law for Set Complement over Union Using a Venn Diagram

De Morgan's Law for the complement of the union of two sets A and B states that: $(A \cup B)' = A' \cap B'$. Use a Venn diagram to prove that De Morgan's Law is true.

Solution

Step 1: First, draw a Venn diagram representing the left side of the equality. The regions of interest are shaded to highlight the sets of interest. $A \cup B$ is shaded on the left, and $(A \cup B)'$ is shaded on the right.

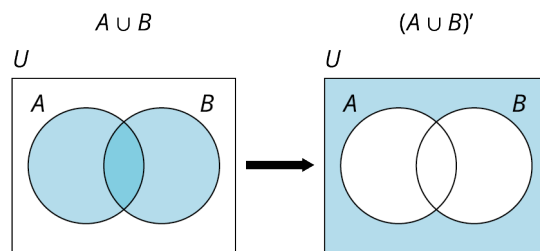


Figure 1.39

Step 2: Next, draw a Venn diagram to represent the right side of the equation. A' is shaded and B' is shaded. Because A' and B' mix to form $A' \cap B'$ is also shaded.

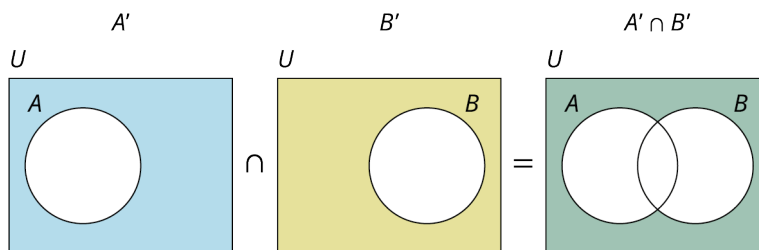


Figure 1.40 Venn diagram of intersection of the complement of two sets.

Step 3: Verify the conclusion. Because the shaded region in the Venn diagram for $(A \cup B)'$ matches the shaded region in the Venn diagram for $A' \cap B'$, the two sides of the equation are equal, and the statement is true. This completes the proof that De Morgan's law is valid.

> YOUR TURN 1.38

1. De Morgan's Law for the complement of the intersection of two sets A and B states that $(A \cap B)' = A' \cup B'$. Use a Venn diagram to prove that De Morgan's Law is true.

Check Your Understanding

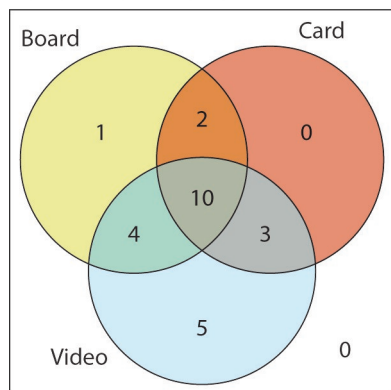
28. When creating a Venn diagram with two or more subsets, you should begin with the region involving the most _____, then work your way from the center outward.
29. To construct a Venn diagram with three subsets, draw and label three circles that overlap in a common _____ region inside the rectangle of the universal set to represent each of the three subsets.
30. In a Venn diagram with three sets, the area where all three sets, A , B , and C overlap is equal to the set _____.
31. When performing set operations with three or more sets, the order of operations is inner most _____ first, then find the _____ of any sets, and finally perform any union or intersection operations that remain.
32. To prove set equality using Venn diagrams, draw a Venn diagram to represent each side of the _____ and then compare the diagrams to determine if they match or not. If they match, the statement is _____, otherwise it is not.



SECTION 1.5 EXERCISES

A gamers club at Baily Middle School consisting of 25 members was surveyed to find out who played board games, card games, or video games. Use the results depicted in the Venn diagram below to answer the following exercises.

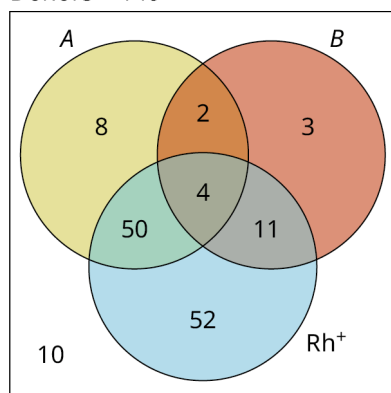
Gamers Club Members = 25



1. How many gamers club members play all three types of games: board games, card games, and video games?
2. How many gamers are in the set $\text{Board} \cap \text{Video}$?
3. If Javier is in the region with a total of three members, what type of games does he play?
4. How many gamers play video games?
5. How many gamers are in the set $\text{Board} \cup \text{Card}$?
6. How many members of the gamers club do not play video games?
7. How many members of this club only play board games?
8. How many members of this club only play video games?
9. How many members of the gamers club play video and card games?
10. How many members of the gamers club are in the set Card' ?

A blood drive at City Honors High School recently collected blood from 140 students, staff, and faculty. Use the results depicted in the Venn diagram below to answer the following exercises.

Donors = 140



11. Blood type AB^+ is the universal acceptor. Of the 140 people who donated at City Honors, how many had blood type AB^+ ?
12. Blood type O^- is the universal donor. Anyone needing a blood transfusion can receive this blood type. How many people who donated blood during this drive had O^- blood?
13. How many people donated with a type A blood factor?
14. How many people donated with a type A and type B blood factor (that is, they had type AB blood).
15. How many donors were O^+ ?
16. How many donors were not Rh^+ ?
17. Opal has blood type A^+ . If she needs to have surgery that requires a blood transfusion, she can accept blood from anyone who does not have a type B blood factor. How many people donated blood during this drive at City Honors that Opal can accept?
18. Find $n(A \cap \text{Rh}^+)$.
19. Find $n(A \cup \text{Rh}^+)$.
20. Find $n(A \cap B \cap \text{Rh}^-)$.

For the following exercises, create a three circle Venn diagram to represent the relationship between the described sets.

21. The number of elements in the universal set, U , is $n(U) = 48$. Sets A , B , and C are subsets of U : $n(A) = 23$, $n(B) = 25$, and $n(C) = 17$. Also, $n(A \cap B) = 15$, $n(B \cap C) = 12$, $n(C \cap A) = 11$, and $n(A \cap B \cap C) = 8$.
22. The number of elements in the universal set, U , is $n(U) = 88$. Sets A , B , and C are subsets of U : $n(A) = 31$, $n(B) = 46$; $n(C) = 33$. Also, $n(A \cap B) = 24$, $n(B \cap C) = 24$, $n(C \cap A) = 26$, and $n(A \cap B \cap C) = 22$.
23. The number of elements in the universal set, U , is $n(U) = 52$. Sets A , B , and C are subsets of U : $n(A) = 23$, $n(B) = 27$, and $n(C) = 29$. Also, $n(A \cap B) = 22$, $n(B \cap C) = 21$, $n(C \cap A) = 19$, and $n(A \cap B \cap C) = 18$.
24. The number of elements in the universal set, U , is $n(U) = 144$. Sets A , B , and C are subsets of U : $n(A) = 36$, $n(B) = 64$, and $n(C) = 81$. Also, $n(A \cap B) = 26$, $n(B \cap C) = 61$, $n(C \cap A) = 29$, and $n(A \cap B \cap C) = 25$.
25. The universal set, U , has a cardinality of 36.
 $n(A) = 12$, $n(B) = 12$, $n(C) = 15$, $n(A \text{ and } B) = 3$, $n(B \text{ and } C) = 4$, $n(C \text{ and } A) = 5$, $n(C \text{ and } A) = 5$, and $n(A \text{ and } B \text{ and } C) = 1$.
26. The universal set, U , has a cardinality of 63. $n(A) = 29$,
 $n(B) = 31$, $n(C) = 41$, $n(A \text{ and } B) = 12$, $n(B \text{ and } C) = 16$, $n(C \text{ and } A) = 18$, and $n(A \text{ and } B \text{ and } C) = 5$.
27. The universal set, U , has a cardinality of 72.
 $n(A) = 32$, $n(B) = 32$, $n(C) = 44$, $n(A \text{ and } B) = 18$, $n(B \text{ and } C) = 22$, $n(C \text{ and } A) = 26$, and $n(A \text{ and } B \text{ and } C) = 14$.
28. The universal set, U , has a cardinality of 81.
 $n(A) = 54$, $n(B) = 41$, $n(C) = 52$, $n(A \text{ and } B) = 32$, $n(B \text{ and } C) = 28$, $n(C \text{ and } A) = 30$, and $n(A \text{ and } B \text{ and } C) = 21$.
29. The anime drawing club at Pratt Institute conducted a survey of its 42 members and found that 23 of them sketched with pastels, 28 used charcoal, and 17 used colored pencils. Of these, 10 club members used all three mediums, 18 used charcoal and pastels, 11 used colored pencils and charcoal, and 12 used colored pencils and pastels. The remaining club members did not use any of these three mediums.
30. A new SUV is selling with three optional packages: a sport package, a tow package, and an entertainment package. A dealership gathered the following data for all 31 of these vehicles sold during the month of July. A total of 18 SUVs included the entertainment package, 11 included the tow package, and 16 included the sport package. Of these, five SUVs included all three packages, seven were sold with both the tow package and sport package, 11 were sold with the entertainment and sport package, and eight were sold with the tow package and entertainment package. The remaining SUVs sold did not include any of these optional packages.

For the following exercises, perform the set operations as indicated on the following sets:

$U = \{\text{red, orange, yellow, green, blue, indigo, violet}\}$, $A = \{\text{red, yellow, blue}\}$, $B = \{\text{orange, green, violet}\}$, and $C = \{\text{red, green, indigo}\}$.

31. Find $(A \cup B) \cap C$.
32. Find $(A \cap C) \cup B$.
33. Find $U \cap (B \cup C)$.
34. Find $(B \cap A) \cap U$.
35. Find $A \cap (B \cap C)'$.
36. Find $A' \cap (B \cup C)$.

For the following exercises, perform the set operations as indicated on the following sets:

$U = \{20, 21, 22, \dots, 29\}$, $A = \{21, 24, 27\}$, $B = \{20, 22, 24, 28\}$, and $C = \{21, 23, 25, 27\}$.

37. Find A and B and C' .
38. Find A' or B or C .
39. Find $(A \text{ or } B)$ and C' .
40. Find $(A \text{ or } B)$ or C' .
41. Find $(A \text{ and } C)$ and B' .
42. Find $(A \text{ or } B)'$ and C .

For the following exercises, use Venn diagrams to prove the following properties of sets:

43. Commutative property for the union of two sets: $A \cup B = B \cup A$.
44. Commutative property for the intersection of two sets: $A \cap B = B \cap A$.
45. Associative property for the intersection of three sets: $(A \cap B) \cap C = A \cap (B \cap C)$.
46. Associative property for the union of three sets: $A \cup (B \cup C) = (A \cup B) \cup C$.
47. Distributive property for set intersection over set union: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
48. Distributive property for set union over set intersection: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.